

A Comparative Benchmark of Semantic, Statistical, and Structural Retrieval Paradigms for Retrieval-Augmented Generation over SEC 10-K Filings

Design and Specification Proposal, COMP702 M.Sc. Project (2025/26)

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Statement of Ethical Compliance: A0

Data Category: A

Participant Category: 0

I confirm that I have read the ethical guidelines and will follow them during this project. This project uses only publicly available, non-personal United States SEC 10-K corporate filings sourced from SEC EDGAR; it involves no human participants and no personal data of any kind.

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1. Project Description

Retrieval-Augmented Generation (RAG) systems answer questions by pairing a retriever that selects relevant evidence from a corpus with a language model that generates a grounded answer (Lewis *et al.*, 2020). Most RAG benchmarks target encyclopaedic or open-domain text; far less is known about how different retrieval paradigms behave on dense, table-heavy financial disclosures such as SEC 10-K filings, where an answer may sit inside a single cell of a multi-row table rather than a sentence of prose (Chen *et al.*, 2021). This project builds and benchmarks three complete RAG pipelines, a semantic (dense vector) retriever, a statistical (BM25) retriever, and a structural (hierarchical summary-tree) retriever, against a 140-query, human-anchored benchmark spanning direct and inferential questions over both narrative text and financial tables, to establish where each paradigm succeeds and where it degrades.

2. Aims and Requirements

2.1 Aims

- Design and implement three architecturally distinct RAG retrieval pipelines (semantic, statistical, structural) sharing one evaluation harness, so that performance differences are attributable to retrieval strategy rather than to generation.
- Construct an adversarially verified, human-anchored benchmark of 140 questions, stratified across four query types (direct/implicit retrieval over text/table content), grounded in real SEC 10-K filings.
- Quantify each pipeline’s retrieval accuracy, answer quality, and computational efficiency, and identify the conditions under which each paradigm degrades.

2.2 Requirements: Essential

- Parse SEC 10-K filings into metadata-enriched, atomic text/table nodes without bisecting tables.
- Implement P1 (dense embeddings with a cross-encoder reranker), P2 (Okapi BM25), and P3 (hierarchical summary-tree) retrievers sharing one answering model.
- Generate and adversarially verify a 140-query benchmark with node-level ground-truth citations.
- Validate an automated LLM judge against human scores before using it to grade the full benchmark.
- Persist all queries, retrieved evidence, and scores in a relational (SQLite) store with crash-safe resume.

2.3 Requirements: Desirable

- Extend the comparison with a metadata pre-filter or hybrid-fusion ablation (reserved future-scope pipelines).
- Visualise per-quadrant performance gaps in a comparison view for the dissertation write-up.

3. Key Literature and Background Reading

Lewis *et al.* (2020) framed answer generation grounded in retrieved evidence as a way of reducing hallucination relative to purely parametric generation, and this Retrieval-Augmented Generation framing motivates the project's overall design: a retriever supplies evidence, and one shared answerer consumes it.

On the statistical side, Robertson and Zaragoza (2009) formalised the probabilistic relevance framework that underlies BM25, which remains a strong, fully deterministic retrieval baseline and forms this project's P2 pipeline. On the semantic side, Karpukhin *et al.* (2020) showed that learned dense bi-encoder retrieval can outperform BM25 by 9-19% absolute on open-domain question answering, while Nogueira and Cho (2019) showed that adding a cross-encoder reranking stage over an initial candidate set materially improves precision; together these findings motivate the project's P1 embed-then-rerank design. The embedding and reranker models used for P1 draw on the BGE family released by Xiao *et al.* (2023).

Sarathi *et al.* (2024) introduced RAPTOR, which recursively clusters and summarises text into a retrievable tree; this is the architectural basis for the project's P3 structural pipeline, although their evaluation, like most RAG literature, is conducted on narrative rather than tabular text.

The domain challenge that motivates this project is well documented: Chen *et al.* (2021) introduced FinQA and showed that even strong pre-trained models fall far short of expert humans at multi-step numerical reasoning over financial tables, which is why the benchmark constructed here explicitly stratifies table-heavy query quadrants. Liu *et al.* (2024) further showed that model accuracy degrades when relevant evidence sits in the middle of a long context, which is the argument for retrieval-based context selection over naive full-document stuffing.

Finally, on evaluation methodology, Zheng *et al.* (2023) showed that a strong LLM judge can reach over 80% agreement with human preference judgements, the same threshold this project adopts as its judge-validation gate before trusting an automated judge to grade the full benchmark.

None of the studies above evaluate all three retrieval paradigms side by side on a single financial-filing benchmark stratified by both query type and content modality (text versus table); this is the specific gap the present project addresses.

4. Development and Implementation Summary

The system is implemented in Python 3, using LlamaIndex as a common retrieval and orchestration backbone so that all three pipelines expose the same retriever interface. Generation, dataset

critique, and judging are performed by open-source instruction models served free of charge via Groq (rate-limited rather than billed, so the implementation has zero recurring spend). Embeddings (BAAI/bge-small-en-v1.5) and reranking (BAAI/bge-reranker-base) run locally on CPU.

ChromaDB provides the local vector store for P1, *rank_bm25* provides the keyword index for P2, and a cached LlamaIndex SummaryIndex provides the structural pipeline for P3. A single SQLite database, run in Write-Ahead Logging mode, is the sole state store; every benchmark run commits immediately, so a crash resumes from the last written row rather than re-spending free-tier quota. Asynchronous worker pools, bounded by a semaphore and wrapped in an exponential-backoff retry layer, keep concurrent API calls within Groq’s free-tier rate limits.

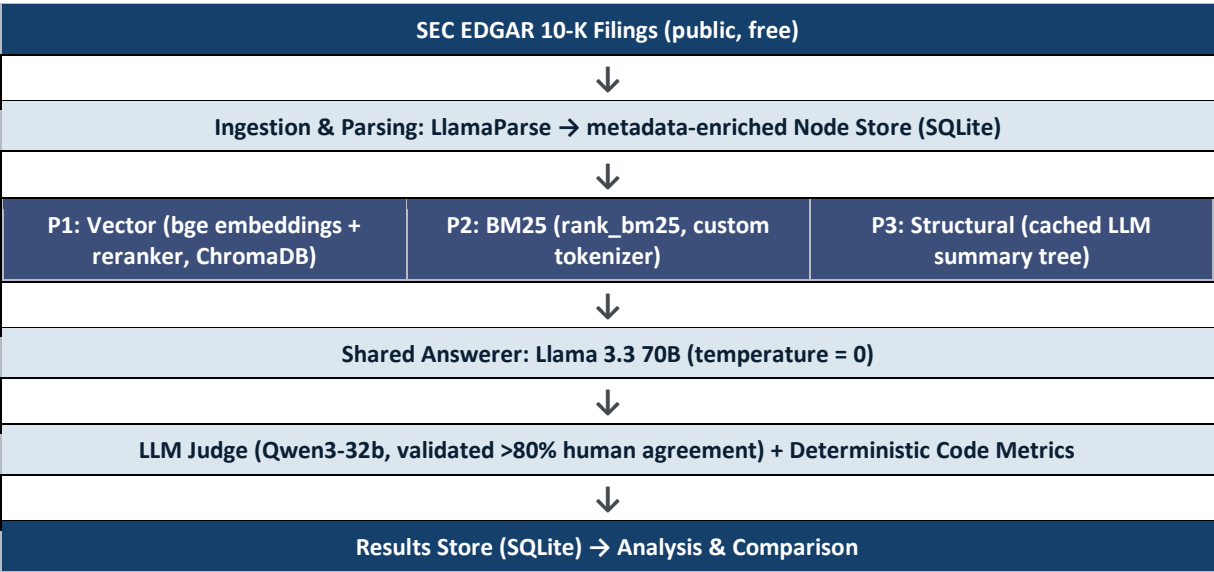


Figure 04.01: High-Level Pipeline Architecture

5. Data Sources

All data is sourced from SEC EDGAR, the United States Securities and Exchange Commission’s public filing system: a corpus of approximately six to nine 10-K annual filings, drawn from two to three companies across two to three fiscal years. These are public corporate disclosures and contain no personal or human-subject data. Filings are parsed via LlamaParse (free tier) into metadata-enriched text and table nodes.

The 140-query evaluation benchmark is itself synthetically generated and adversarially verified from this corpus by two independent open-source language models acting as a cross-family generator and critic, with no third-party data collection and no participant data involved at any stage. This satisfies Data Category A0.

Key point. Every recurring component in this pipeline runs on a free tier or local CPU, so the expected spend for the full benchmark is \$0.00; the only binding constraint is Groq’s daily token throughput, not budget.

6. Testing and Evaluation

The system is evaluated across three independent pillars. **Retrieval metrics** (Precision@K, Recall@K, Evidence Hit Rate) are computed deterministically against node-ID ground truth. **Answer-quality metrics** use an LLM-as-judge score (1-10) as the primary measure, with token-level F1 and numeric-tolerant Exact Match as secondary lexical checks, plus a code-level Citation Audit that flags answers which are correct only by coincidence, where the cited evidence does not match the ground truth. **Efficiency metrics** cover latency, token consumption, and one-time index-build cost per pipeline.

The automated judge is not trusted by default. It must first clear a hard validation gate, reaching greater than 80% agreement with the researcher's own hand-scored outputs on a disjoint validation set, before it is permitted to grade the full benchmark, mirroring the agreement threshold reported for LLM judges by Zheng *et al.* (2023). Every script that calls a language model is first run end-to-end under a hardcoded three-item local throttle, to catch bugs before any free-tier quota is spent at scale.

Key point. The judge is gated, not trusted by default: it must clear 80% agreement with hand-scored human judgements on a disjoint validation set before it is allowed to grade the full 900-run benchmark.

7. Project Ethics and Human Participants

This project involves no human participants and no personal data; all source material is SEC 10-K filings, which are public corporate disclosures (Data Category A0). The two manual steps in the methodology, hand-labelling 20 teaching exemplars and hand-scoring 60 judge-validation outputs, are carried out solely by the researcher as part of constructing the evaluation harness itself, rather than as a study involving external participants, and therefore fall outside the scope of human-participant ethics review. No further ethical approval beyond this statement is required for the activity planned within this assessment.

8. BCS Project Criteria

- Systematic understanding: a cross-paradigm RAG comparison grounded in current literature (Lewis et al., 2020; Karpukhin et al., 2020; Sarthi et al., 2024), exposing paradigm limits not yet documented for SEC-filing data.
- Comprehensive technique understanding: dense retrieval, sparse retrieval, hierarchical summarisation, reranking, and LLM-based evaluation are all designed and built within a single project.

- **Originality:** the cross-family generator/critic adversarial verification pipeline and the disjoint three-set (PQ/GQ/JEQ) evaluation design are original contributions to constructing a RAG benchmark without manual annotation at scale.
- **Sound judgement under incomplete information:** deliberate, documented scoping decisions, such as dropping a metadata pre-filter and accepting a token-per-day-bound runtime, demonstrate judgement under real resource constraints.
- **Self-direction:** a one person, ten week build plan run entirely on free-tier infrastructure with no managed budget, together with critical self-evaluation of the project’s own statistical limitations, demonstrates autonomous planning.

9. UI/UX Mockup

This is a research and benchmarking system rather than an end-user application, so its principal “interface” is the results comparison view produced at the end of the run and consulted directly in the dissertation analysis. A wireframe sketch of this planned output is shown below.

Pipeline	K	Precision@K	Recall@K	Judge Score	Latency (s)
P1_vector	5	0.62	0.81	8.1	1.84
P2_bm25	5	0.54	0.73	7.0	0.91
P3_structural	5	0.41	0.66	6.4	1.10

Figure 09.01: Results Comparison View (Wireframe, illustrative placeholder values, not final results)

10. Project Plan

The build is scoped to a one person, ten-week schedule, with active development substantially complete by Week 7 and the final three weeks overlapping the full benchmark run with results analysis and dissertation write-up.

Phase	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
1. Environment & Infra										
2. Ingestion & Parsing										
3. P3 Summary-Index Build										
4. Dataset Gen & Verification										
5. Pipeline Impl. (P1/P2/P3)										
6. Judge Build & Validation Gate										
7. Full Benchmark Execution										
8. Results Analysis & Write-up										

Table 10.01: 10-Week Project Gantt Chart

11. Risks and Contingency Plans

Risk	Contingency	Likelihood	Impact
Groq free-tier token-per-day limit reached mid-run	Exponential backoff with jitter, multi-day paced execution with crash-safe resume, prompt caching, fallback to the paid Developer tier if the schedule slips	Medium	Medium
LlamaParse fragments or bisects a financial table	Randomised 20-section manual parsing audit; re-tune parser settings or discard and re-parse the affected filing	Medium	High
Automated LLM judge fails the 80% human-agreement gate	Tune the scoring rubric and/or swap the judge to a different model family; re-run the validation gate before proceeding	Medium	High
Breaking API changes in a fast-moving dependency (LlamaIndex)	Pin exact package versions at project start; consult current documentation before any upgrade	Low	Medium
Concurrent writes corrupt the SQLite results store mid-run	Write-Ahead Logging (WAL) mode; crash-safe resume keyed on a unique constraint; commit after every individual run	Low	High
One person, ten-week timeline slips	Throttle-first testing on every phase before full-scale release; background-paced benchmark execution; buffer reserved across the final three weeks	Medium	Medium

Table 11.01: Risk Management Plan

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